

Technical Document #764: Database Systems - Comprehensive Overview

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Relational databases organize data into tables with rows and columns. SQL is used to query and manipulate structured data.

NoSQL databases handle unstructured and semi-structured data. Document stores, key-value stores, and graph databases are common NoSQL types.

ACID properties (Atomicity, Consistency, Isolation, Durability) ensure reliable transaction processing in databases.

Database replication copies data across multiple servers. It improves availability and provides fault tolerance.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Key Takeaways:

- Database indexing improves query performance by creating data structures that allow faster data retrieval.
- Database migration tools manage schema changes and data transformations.
- Database monitoring tracks performance metrics and identifies bottlenecks.